

ManiSkill3 Setup Guide

AlohaMini Robot Integration

Technical Documentation

AlohaMini Project

January 14, 2026

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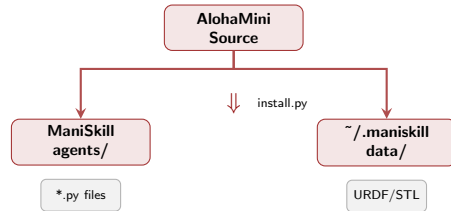
Setup Overview

Purpose:

Integrate AlohaMini robot into ManiSkill3 simulation framework.

Key Components:

- `install.py` - Installation script
- `setup.py` - Package configuration
- Agent files (*.py)
- URDF & mesh assets



Prerequisites

System Requirements:

- Python $\geq$ 3.8
- CUDA (for GPU acceleration)
- Linux / macOS / Windows

Package Dependencies:

- mani-skill $\geq$ 3.0.0b0
- numpy
- pygame (optional, for keyboard)

Installation Commands:

```
1 # Create virtual environment
2 python -m venv venv
3 source venv/bin/activate
4
5 # Install ManiSkill3
6 pip install mani-skill
7
8 # Verify installation
9 python -c "import mani_skill; print(
    mani_skill.__version__)"
```

setup.py Configuration

```
1 from setuptools import setup, find_packages
2
3 setup(
4     name="aloha_mini_maniskill",
5     version="0.1.0",
6     description="AlohaMini dual-arm mobile robot for ManiSkill3",
7     packages=find_packages(),
8     package_data={
9         "aloha_mini": ["*.urdf", "meshes/*.STL"],
10    },
11    install_requires=[
12        "mani-skill>=3.0.0b0",
13        "numpy",
14    ],
15    extras_require={
16        "dev": ["pygame"], # For keyboard control
17    },
18    python_requires=">=3.8",
19 )
```

Installation Process

4-Step Process:

1 Agent Files

Copy *.py to ManiSkill agents

2 URDF & Meshes

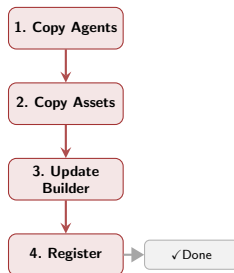
Copy to ~/.maniskill/data/

3 Scene Builder

Update ReplicaCAD integration

4 Registration

Add import to __init__.py



install.py - Path Discovery

```
1 def find_maniskill_path():
2     """Find the ManiSkill installation path."""
3     try:
4         import mani_skill
5         return Path(mani_skill.__file__).parent
6     except ImportError:
7         print("Error: ManiSkill not installed.")
8         sys.exit(1)
9
10 def get_maniskill_data_dir():
11     """Get the ManiSkill data directory."""
12     return Path.home() / ".maniskill" / "data"
```

Key Insight

ManiSkill stores robot assets in `~/.maniskill/data/robots/`

install.py - Copy Files

```
1 def install():
2     script_dir = Path(__file__).parent.resolve()
3     maniskill_path = find_maniskill_path()
4     data_dir = get_maniskill_data_dir()
5
6     # 1. Install agent files
7     agent_src = script_dir / "agents" / "aloha_mini"
8     agent_dst = maniskill_path / "agents" / "robots" / "aloha_mini"
9     agent_dst.mkdir(parents=True, exist_ok=True)
10    for f in agent_src.glob("*.py"):
11        shutil.copy2(f, agent_dst / f.name)
12
13    # 2. Install URDF and mesh files
14    asset_src = script_dir / "assets" / "robots" / "aloha_mini"
15    asset_dst = data_dir / "robots" / "aloha_mini"
16    shutil.copytree(asset_src, asset_dst)
```

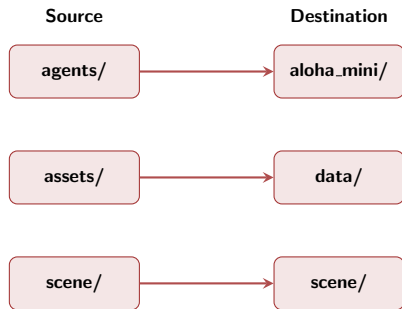

Directory Mapping

Source (AlohaMini):

- `maniskill/agents/aloha_mini/*.py`
- `maniskill/assets/robots/aloha_mini/`
- `maniskill/scene_builder/`

Destination (ManiSkill):

- `mani_skill/agents/robots/aloha_mini/`
- `~/.maniskill/data/robots/aloha_mini/`
- `mani_skill/utils/scene_builder/`



Asset Directory Structure

```
~/maniskill/data/robots/aloha_mini/  
+-- aloha_mini_so100.urdf  
+-- so100_meshes/  
    +-- Base.stl  
    +-- Shoulder.stl  
    +-- Upper_Arm.stl  
    +-- Forearm.stl  
    +-- Wrist.stl  
    +-- Fixed_Jaw.stl  
    +-- Moving_Jaw.stl
```

URDF Configuration:

- Robot model definition
- Link/joint hierarchy
- Collision & visual meshes
- Inertia properties

Mesh Files:

- STL format (binary)
- Exported from CAD
- Used for rendering & collision

@register_agent Decorator

```
1 from mani_skill.agents.base_agent import BaseAgent
2 from mani_skill.agents.registration import register_agent
3
4 @register_agent()
5 class AlohaMiniSO100V2(AlohaMiniBaseAgent):
6     uid = "aloha_mini_so100_v2"
7     urdf_path = "robots/aloha_mini/aloha_mini_so100.urdf"
8
9     # Joint configuration
10    arm_joint_names = [
11        "left_shoulder_pan", "left_shoulder_lift",
12        "left_elbow_flex", "left_wrist_flex",
13        "left_wrist_roll", "left_gripper",
14        # ... right arm joints
15    ]
```

Register in `__init__.py`

```
1 # mani_skill/agents/robots/__init__.py
2
3 # Add import for AlohaMini
4 from .aloha_mini import AlohaMiniS0100V2
5
6 # install.py automatically adds this import:
7 robots_init = maniskill_path / "agents" / "robots" / "__init__.py"
8 if robots_init.exists():
9     content = robots_init.read_text()
10     if "aloha_mini" not in content:
11         new_import = 'from .aloha_mini import AlohaMiniS0100V2'
12         content = f"{new_import}\n{content}"
13     robots_init.write_text(content)
```

Result

After installation: `from mani_skill.agents.robots import AlohaMiniS0100V2`

ReplicaCAD Scene Builder

Purpose:

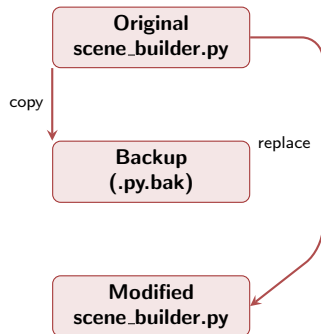
Enable AlohaMini in ReplicaCAD environments.

Modified File:

[mani_skill/utils/scene_builder/replicacad/scene_builder.py](#)

Backup:

Original saved as `.py.bak`



Scene Builder Update

```
1 # install.py - Scene builder update
2 scene_builder_src = script_dir / "scene_builder" / "replicacad" / "scene_builder.py"
3 scene_builder_dst = maniskill_path / "utils" / "scene_builder" / "replicacad" / "
    scene_builder.py"
4
5 if scene_builder_src.exists() and scene_builder_dst.exists():
6     # Backup original
7     backup = scene_builder_dst.with_suffix(".py.bak")
8     if not backup.exists():
9         shutil.copy2(scene_builder_dst, backup)
10        print(f"Backed up original to {backup.name}")
11
12    # Replace with modified version
13    shutil.copy2(scene_builder_src, scene_builder_dst)
14    print(f"Updated scene_builder.py")
```

Verification - Import Check

Step 1: Verify Python Import

```
1 python -c "from mani_skill.agents.robots import aloha_mini; print('OK')"
```

Expected Output:

```
OK
```

If ImportError

ManiSkill not installed or agent not registered.

Fix: `pip install mani-skill && python install.py`

Verification - Run Demo

Step 2: Test Teleoperation Demo

```
1 cd maniskill/teleop  
2 python demo_teleop.py --render
```

Expected Behavior:

- ManiSkill3 viewer window opens
- AlohaMini robot visible in scene
- Keyboard controls functional (WASD, arrows)

Tip

Use `--help` flag to see all available options.

Verification - File Check

Step 3: Verify Installed Files

Agent Files:

```
1 ls $(python -c "  
2 import mani_skill  
3 print(mani_skill.__file__.replace(  
4     '__init__.py',  
5     'agents/robots/aloha_mini'))")
```

Expected:

- `__init__.py`
- `base_agent.py`
- `aloha_mini_so100_v2.py`

Asset Files:

```
1 ls ~/.maniskill/data/robots/aloha_mini/
```

Expected:

- `aloha_mini_so100.urdf`
- `so100_meshes/`

Troubleshooting - Common Errors

Error	Cause	Solution
<code>ImportError</code>	ManiSkill not installed	<code>pip install mani-skill</code>
<code>FileNotFoundError</code>	URDF/mesh missing	Re-run <code>python install.py</code>
<code>ModuleNotFoundError</code>	Agent not registered	Check <code>__init__.py</code> imports

Debugging Tip

Run with verbose flag for detailed output: `python install.py --verbose`

Troubleshooting - Quick Fixes

Complete Reinstall:

```
python install.py --uninstall  
python install.py
```

Check ManiSkill Version:

```
python -c "import mani_skill; print(  
    mani_skill.__version__)"
```

Verify Paths:

```
python -c "import mani_skill; print(  
    mani_skill.__file__)"
```

Check Data Directory:

```
ls -la ~/.maniskill/data/robots/
```

Uninstallation - Command

Remove AlohaMini from ManiSkill

```
python install.py --uninstall
```

What Gets Removed:

- **Agent Files:** `mani_skill/agents/robots/aloha_mini/`
- **Asset Files:** `~/.maniskill/data/robots/aloha_mini/`
- **Scene Builder:** Restores original from `.py.bak` backup

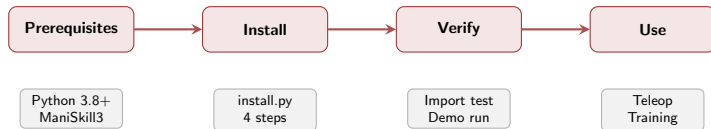
Safe Operation

Original `scene_builder.py` is restored from backup created during installation.

Uninstallation - Code

```
1 def uninstall():
2     maniskill_path = find_maniskill_path()
3     data_dir = get_maniskill_data_dir()
4
5     # Remove agent files
6     agent_dst = maniskill_path / "agents" / "robots" / "aloha_mini"
7     if agent_dst.exists():
8         shutil.rmtree(agent_dst)
9
10    # Remove asset files
11    asset_dst = data_dir / "robots" / "aloha_mini"
12    if asset_dst.exists():
13        shutil.rmtree(asset_dst)
14
15    # Restore scene builder backup
16    backup = scene_builder_dst.with_suffix(".py.bak")
17    if backup.exists():
18        shutil.copy2(backup, scene_builder_dst)
19        backup.unlink() # Remove backup file
```

Summary



Quick Start

```
pip install mani-skill && python install.py && python demo_teleop.py
```

Setup Complete

ManiSkill3 + AlohaMini Integration

Robot UID: `aloha_mini_so100_v2`